

ME 563 – Intermediate Fluid Dynamics

Instructor: Prof. Wenxiao Pan
Office Hours: by appointment
Office: ME 2242
Email: wpan9@wisc.edu

Grader: Anirudh Reddy Bendaram
Office: ME2042
Email: bendaram@wisc.edu
Office Hours: by appointment

Lectures: MW, 4:00PM - 5:15PM
Room: MECH ENGR 1152

Course Homepage: <https://canvas.wisc.edu/courses/520345/assignments> (sign in through UW Canvas)

Synopsis:

This course is intended for first-year graduate engineering students who have taken at least one undergraduate fluid mechanics course in engineering (equivalent to ME 363) but may be also suitable for advanced undergraduate students. This course focuses on developing a detailed understanding of the underlying physics and mathematics of fluid dynamics. This level of understanding is needed to develop computational fluid dynamics models, to perform detailed analysis of spatially- and temporally-resolved fluid-dynamics data, and to research and analyze fundamental and applied fluid-dynamics problems. The course first reviews the relevant mathematics and develops understanding of the kinematics of fluid motion and governing equations. These tools are then applied to analyze inviscid and potential flows, followed by applications to canonical laminar flows of engineering interest.

Objectives:

- Improve physical and mathematical understanding of fluid flows.
- Learn the basic analysis methods and apply fluid dynamics to applications at a more fundamental level than commonly pursued in undergraduate classes.
- Apply the governing equations of fluid dynamics to flows of engineering interest.
- Understand and use advanced analysis techniques for solving fluids problems.
- Obtain an understanding of advanced fluid dynamics topics such as vortex dynamics and boundary layer theory.

Class Structure:

There are two lectures per week that cover the detailed concepts and theory with use of quantitative examples when possible. There are two midterm exams and a final exam (*not cumulative*).

Course Policies:

- The homework and exams will be graded on completeness, correctness and style. Failure to present your results in a concise, easy to follow manner will result in deductions. Sufficient work needs to be shown to see how the results are obtained.
- *Any student who attempts to gain unfair advantage over other students on Academic Integrity will fail this course.*

Homework:

Homework assignments will be due approximately every week (see Course Schedule) and available on the course Canvas website. Electronic submission (in PDF format) is required and through Canvas. Homework printouts or e-mail attachments will not be accepted. The homework will be graded on both correctness and style. Failure to present your results in a concise, easy to follow manner will result in deductions. Sufficient work needs to be shown to see how the result was derived. Each person must turn in their own homework assignment and the work must be all their own (see Academic Integrity Statement on Page 4), but you are allowed to discuss the homework with others while you are working on it. **Late homework will not be accepted for any reason.** Your lowest homework grade will be dropped. Save this drop for emergencies!

Exam:

There will be **two** in-class midterm exams given on the dates shown on the schedule. The exams will be administered during class and will cover recent material unless otherwise stated. The final exam is scheduled on **Dec 11 (Friday) at 5:05PM - 7:05PM**. The final exam is not cumulative but will be 1.5x in length compared to the in-class midterms. All exams are open book, and notes and a calculator are also allowed. However, **the use of the internet or AI to search for solutions during the exam is strictly prohibited!** Any student who violates this rule will be considered to have gained an unfair academic advantage and will be subject to academic integrity violations, resulting in failure of the course.

- Make-up exams may be given only in extreme cases for legitimate documentable reasons. If for some legitimate reason, you cannot make one of the exams, contact the instructor at least one week before the exam, and the make-up exam must be taken *before* the scheduled exam date.
- Do NOT plan to leave town before the final exam.

Evaluation and Grading:

Homework	30%
Midterm 1	20%
Midterm 2	20%
Final Exam	30%

Grades in this course will **NOT** be assigned on a curve!

The grading scale will be: 93-100 (A); 88-92 (AB); 83-87 (B); 78-82 (BC); 70-77 (C); 60-69 (D); below 60 (F).

Course Prerequisites:

An undergraduate engineering course in fluid dynamics (equivalent to ME 363) is required along with strong math background and ability to program in Matlab/Python or similar environment.

Recommended Textbook:

K. Kundu, I. M. Cohen, and D. R. Dowling, Fluid Mechanics, 5th or 6th ed. Amsterdam: Academic Press.

Other References:

A list of additional resources that contain information relevant to the topics of this course:

1. Acheson, D.J., Elementary Fluid Dynamics, Clarendon Press, Oxford, 1990.
2. Aris, R., Vectors, Tensors, and the Basic Equations of Fluid Mechanics, Dover Publications, 1962.
3. Batchelor, G.K., An Introduction to Fluid Dynamics, Cambridge University Press, 1967.
4. Panton, R. L., Incompressible Flow, John Wiley and Sons, 3rd ed., 2005.
5. Pozrikidis, Fluid Dynamics, Springer, 2001
6. Tritton, D.J., Physical Fluid Dynamics, Von Nostrand Reinhold, 1988.
7. White, F.M., Viscous Fluid Flow, McGraw Hill, 2nd edition, 1991.

Course Schedule:

#	Date	Day	Topic	Reading*	HW Due/Exam
1	2-Sep	W	Introduction, Fundamentals	§1.1-1.11	
2	9-Sep	W	Dimensional Analysis, Vector	4.11, §2.1-2.9	
3	14-Sep	M	Vector Calculus	§2.10-2.13, Aris chap 2&3	
4	16-Sep	W	Tensor	§2.10-2.13, Aris chap 2&3	HW1
5	21-Sep	M	Fluid Kinematics I	§3.1, 3.4	
6	23-Sep	W	Fluid Kinematics II	§3.5, 3.6	HW2
7	28-Sep	M	Reynold's Transport Theorem/Conservation Laws-Mass	§3.2, 4.1-4.2	
8	30-Sep	W	Pathlines, Streamlines & Stream Function	§3.2-3.3, 4.3	HW3
9	5-Oct	M	Conservation Laws-Momentum, Integral	§4.4, 4.7, 4.9	
10	7-Oct	W	Conservation Laws-Momentum, Differential	§4.4	HW4
	12-Oct	M	In-Class Midterm 1		Midterm 1
11	14-Oct	W	Constitutive Relations, Navier-Stokes (NS) Equations	§4.5, 4.6	
12	19-Oct	M	Conservation Laws-Energy	§4.8	
13	21-Oct	W	Bernoulli Equation	§4.9	HW5
14	26-Oct	M	Vorticity Dynamics	§5.1-5.4	
15	28-Oct	W	Vorticity Dynamics	§5.5-5.7	HW6
16	2-Nov	M	Vorticity Dynamics/Potential Flow	§5.8, §6(7).1	
17	4-Nov	W	Potential Flow	§6(7).2-6(7).3	HW7
18	9-Nov	M	In-Class Midterm 2		Midterm 2
	11-Nov	W	Potential Flow	§6(7).4	
19	16-Nov	M	Potential Flow	§6(7).5	
20	18-Nov	W	Laminar Flow- Poiseuille, Rotating Concentric Cylinders	§8(9).1-8(9).2	HW8
21	23-Nov	M	Laminar Flow Between Concentric Rotating Cylinders (cont'd) (Recorded Lecture)	§8(9).1-8(9).2	
	25-Nov	W	No Class		
22	30-Nov	M	Laminar Flow-Hydrodynamic Lubrication	§8(9).3, Panton Chap 22	
23	2-Dec	W	Laminar Flow-Stoke's 1 st and 2 nd Problems	§8(9).4-8(9).5	HW9
24	7-Dec	M	Laminar Boundary Layers	§9(10).1-9(10).7	
25	9-Dec	W	Laminar Free Shear Layers	§9(10).10	HW10
	11-Dec	F	Final Exam	5:05PM - 7:05PM	Final Exam

* Chapter and section numbers refer to Kundu, Cohen, and Dowling, Fluid Mechanics, 5th or 6th ed., 2012, 2016. Chapter numbers in parentheses are for the 6th ed.

Academic Integrity Statement

By virtue of enrollment, each student agrees to uphold the high academic standards of the University of Wisconsin-Madison; academic misconduct is behavior that negatively impacts the integrity of the institution. Cheating, fabrication, plagiarism, unauthorized collaboration, and helping others commit these previously listed acts are examples of misconduct which may result in disciplinary action. Examples of disciplinary action include, but is not limited to, failure on the assignment/course, written reprimand, disciplinary probation, suspension, or expulsion.

Accommodations for Students with Disabilities Statement

The University of Wisconsin-Madison supports the right of all enrolled students to a full and equal educational opportunity. The Americans with Disabilities Act (ADA), Wisconsin State Statute (36.12), and UW-Madison policy (Faculty Document 1071) require that students with disabilities be reasonably accommodated in instruction and campus life. Reasonable accommodations for students with disabilities is a shared faculty and student responsibility. Students are expected to inform Prof. Pan of their need for instructional accommodations by the end of the third week of the semester, or as soon as possible after a disability has been incurred or recognized. Prof. Pan will work either directly with the student or in coordination with the McBurney Center to identify and provide reasonable instructional accommodations. Disability information, including instructional accommodations as part of a student's educational record, is confidential and protected under FERPA. (See: [McBurney Disability Resource Center](#))

Religious Observances

Students must notify the instructor within the **first two weeks of class** of the specific days or dates on which you request relief due to religious observances. Make-ups may be scheduled before or after the regularly scheduled requirements.

Use of Lecture Materials and Recordings Statement

Lecture materials and recordings for this course are protected intellectual property at UW-Madison. Students in this course may use the materials and recordings for their personal use related to participation in this class. Students may also take notes solely for their personal use. If a lecture is not already recorded, students are not authorized to record the instructor's lectures without the instructor's permission unless a student is considered by the university to be a qualified student with a disability requiring accommodation. [Regent Policy Document 4-1] Students may not copy or have lecture materials and recordings outside of class, including posting on internet sites or selling to commercial entities. Students are also prohibited from providing or selling their personal notes to anyone else or being paid for taking notes by any person or commercial firm without the instructor's express written permission. Unauthorized use of these copyrighted lecture materials and recordings constitutes copyright infringement and may be addressed under the university's policies, UWS Chapters 14 and 17, governing student academic and non-academic misconduct.